BVG Accessibility in Berlin: Service Gap Analysis

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Summary

Almost every Berlin resident lives close to a BVG stop: 95.9% are within 400 m and 99.85% within 800 m. About 162,000 residents (4.1%) live farther than 400 m away. A set of 280 priority gaps, each at least 1 ha with at least 50 persons per hectare, holds about 59,900 of them, or 37% of everyone beyond 400 m. The analysis covers BVG stops only and uses straight-line distances.

Data and method

The analysis combines three open datasets in QGIS, all processed in EPSG:25833 (ETRS89 / UTM 33N), a metric system suited to Berlin.

- **BVG stops:** from the VBB open GTFS feed. BVG routes were linked to their trips, stop times and stops, giving 7,637 unique BVG-served stops, of which 7,332 lie inside Berlin. Each stop is tagged with its modes (U-Bahn, bus, tram, ferry).
- **City boundary:** ALKIS Berlin Landesgrenze (Senatsverwaltung für Stadtentwicklung, Bauen und Wohnen).
- **Population:** Einwohnerdichte 2025 from the Umweltatlas, in persons per hectare for each block. Residents per block = density × block area, which sums to about 3.91 million for Berlin.

Stops inside Berlin were buffered by 400 m and 800 m and each buffer set was merged into one area. Residents covered = density × the area of each block that falls inside the buffer. Gap areas are Berlin minus the 400 m buffer. Priority gaps are gap pieces of at least 1 ha with a density of at least 50 persons per hectare.

Results

Berlin is almost fully covered: 3,750,666 of about 3,911,453 residents live within 400 m of a BVG stop, and 3,905,652 within 800 m.

Distance to nearest BVG stop	Residents covered	Share of Berlin
400 m	3,750,666	95.9%
800 m	3,905,652	99.85%

About 162,000 residents (4.1%) live more than 400 m from a stop, and only about 5,800 (0.15%) more than 800 m. Subtracting the covered figure from the total gives about 160,800 at 400 m; the small difference comes from calculating the two overlays separately.

By area the picture differs: roughly 28,000 ha, around 31% of Berlin's territory, lies more than 400 m from a stop. On the map these are mostly forests and lakes such as Grunewald, Spandauer Forst, Tegeler Forst and the Müggelsee and Köpenick forests, where few people live.

The 280 priority gaps hold about 59,900 residents, 37% of the people beyond 400 m. They appear mainly along the edges of built-up areas, with a few between inner districts.

Example sites

Five of the largest priority gaps by estimated residents are residential areas just above the 50 persons per hectare threshold, between 50.7 and 54.4.

Site	Density (persons/ha)	Nearby streets or landmarks
Krusauer Straße	51.7	Kettinger Straße, Raabestraße
Pappelweg	50.74	Borders Spandauer Forst
Rupertweg	54.31	Jean-Calas-Weg
Adolfstraße	54.4	Ribbeckweg, Kistostraße
Dahlemer Weg	51.86	Robert-W.-Kempner-Straße, Curtiusstraße

Limitations

The results show where access is weakest, not how good service is. Read them with these caveats:

- **Straight-line distance:** circles overestimate access, because real walking routes are longer and rail lines, rivers and motorways block direct paths.
- **Stops, not service:** frequency, operating hours and reliability are not considered, so a stop with a few trips a day counts like a U-Bahn station.
- **BVG only:** S-Bahn and regional rail run by other operators are not counted, so some gaps may be served by them.
- **Edge effect:** stops were clipped to Berlin before buffering, so stops just across the city border are ignored. Gaps along the boundary, including some of the example sites, may therefore be overstated.
- **Averaged population:** density is a block average, spread evenly inside each block, then cut by circles. Resident counts are estimates.
- **Single snapshot:** one GTFS feed and the 2025 population data. Rail-replacement buses are counted as ordinary bus service.

Next steps

The 280 priority gaps are candidates for review, not confirmed service failures. Four steps would turn them into findings:

1. **Remove the edge effect:** rerun the buffers with stops from a 400 m margin outside the city boundary, then check which boundary gaps remain.
2. **Measure real walking distance:** replace the circles with street-network routes from OpenStreetMap for the remaining gaps.
3. **Add service quality:** count departures per hour from the GTFS stop times and include S-Bahn and regional stops from the same VBB feed, so that "covered" means a usable service.
4. **Rank and inspect:** order the gaps by residents affected and check the largest on site, such as the five example streets above, to judge whether a stop change, an extension or a feeder service is worth considering.